The BGM Breakthrough

11.1 The idea, stated in one sentence

Model the simple forward rates the market actually quotes — $L_i(t) := L(t; T_i, T_{i+1})$ for each tenor date T_i on a fixed grid — *directly*, as lognormal processes each under their own natural forward measure, and derive the drift each one needs under any other measure from the same no-arbitrage machinery HJM used, applied this time to genuinely observable, tradable rates.

This idea was published independently and almost simultaneously by two groups in 1997: Alan Brace, Dariusz Gatarek and Marek Musiela (giving the model its most common name, BGM), and Kristian Miltersen, Klaus Sandmann and Dieter Sondermann. Both papers made the same core observation, and both are properly credited as the origin of what the market now calls either the **BGM model** or, more descriptively, the **LIBOR Market Model (LMM)**.

Key result: what BGM models

BGM fixes a discrete tenor structure $0 = T_0 < T_1 < \cdots < T_N$ (e.g. quarterly dates out to 30 years) and models each forward rate $L_i(t) = L(t; T_i, T_{i+1})$, for $t \leq T_i$, as a stochastic process. The central postulate is that, *under its own T_{i+1}-forward measure $\mathbb{Q}^{i+1}$*, each $L_i(t)$ is a driftless lognormal martingale:

$$\mathrm{d}L_i(t) = \sigma_i(t)\, L_i(t)\, \mathrm{d}W_i^{i+1}(t)$$

where $\sigma_i(t)$ is L_i's own instantaneous volatility and W_i^{i+1} is a Brownian motion under $\mathbb{Q}^{i+1}$. This is exactly the assumption Chapter 6 used, freely and without proof, to derive Black's caplet formula — and that is not a coincidence. It is the entire point of the construction.

11.2 Why this recovers Black's formula exactly

This is worth stating as its own result, because it is the property that made BGM immediately commercially compelling, in a way no HJM implementation or short-rate model had managed before it.

Key result: BGM reproduces Black's caplet formula by construction

Because $L_i(t)$ is postulated, directly, to be driftless lognormal under $\mathbb{Q}^{i+1}$ — the exact measure under which the caplet price (Chapter 6) is a discounted expectation — pricing a caplet on L_i under BGM *is* evaluating Black's formula, with σ_i playing the role of the Black volatility. There is no approximation and no separate derivation required: the model's most basic building block is engineered, by design, to price the single most liquid rates option exactly as the market already prices it.

This single fact is what separates BGM from every model in Chapters 9 and 10. A Hull–White or generic HJM model can be *calibrated* to match a given set of caplet prices, more or less well, through some translation procedure; a BGM model matches them *identically*, for any choice of $\sigma_i(t)$ whatsoever, purely as a structural consequence of the model's own definition.

Calibration, once BGM exists, stops being about "can I hit these caplet prices at all" and becomes entirely about "what volatility and correlation *parameters* best explain the observed level, term structure, and joint behaviour of the whole surface" — a much better-posed and more tractable question, and the one Part V answers in full.

Worked example 11.1 — BGM's caplet price is not new arithmetic.
Take the caplet from Worked Example 6.1: $F = 4.10\%$, $K = 4.00\%$, $T_1 = 1.5$, $\tau = 0.25$, $P(0, T_2) = 0.9702$, $\sigma_n = 90\,\text{bp/yr}$, notional \$10mm, priced at \$11,920 using Bachelier's formula directly.
Step 1 — identify the BGM object. In BGM notation, this caplet's underlying is $L_i(t)$ for the tenor date $T_i = 1.5$, with initial value $L_i(0) = F = 4.10\%$ and (normal) instantaneous volatility calibrated so that $\sigma_{n,i} = 90\,\text{bp/yr}$ matches the quoted market vol for this specific caplet.
Step 2 — price it. Under BGM's own postulated dynamics for $L_i(t)$ (normal version, for consistency with Chapter 6's choice), the caplet price is, by construction, exactly the Bachelier formula applied with $F = L_i(0)$, $\sigma_n = \sigma_{n,i}$ — i.e. exactly \$11,920 again, with no new calculation required.
Interpretation. This is intentionally an anticlimactic worked example: the point being made is that pricing a single caplet under BGM introduces nothing new relative to Chapter 6 — it is, precisely, Chapter 6. BGM's real content, and the reason the rest of this book exists, is in what it lets you do *beyond* a single caplet: price a swaption, a Bermudan, a spread option, anything that depends on more than one forward rate jointly, using a single, internally consistent, jointly-calibrated model rather than a separate ad-hoc formula for every product type.

11.3 The catch: one rate's nice measure is another's dirty one

The postulate above only says $L_i(t)$ is driftless *under its own* measure $\mathbb{Q}^{i+1}$. But a real derivative — a swaption, an exotic, even a simple two-caplet cap when both legs are simulated together — needs every relevant forward rate evolved *simultaneously, under one common measure*. The moment $L_i(t)$ is looked at under any measure other than its own $\mathbb{Q}^{i+1}$, it picks up a drift, and that drift is not zero, not simple, and not even deterministic — it depends on the levels and correlations of every other forward rate in the model.

PITFALL

It is a natural but incorrect first guess that "if every L_i is individually driftless under its own measure, then evolving all of them together must also be driftless." It is not. Measures are specific to a single numeraire; there is exactly one measure under which L_i is driftless, and every other forward rate in the model has a *different* own-measure. Simulating the whole curve requires picking *one* common measure to work under, and every rate except (at most) one will carry a nonzero, state-dependent drift under that common choice. Deriving exactly what that drift must be, for an arbitrary choice of common measure, is the central technical content of Chapter 13, and getting it wrong is the single most consequential bug possible in a BGM Monte Carlo engine — it silently reintroduces arbitrage into simulated prices.

ON THE DESK

This is the trade-off practitioners internalise quickly: BGM is *easy* to calibrate, because its basic building blocks are, by construction, exactly the market's own pricing formulas. BGM is *hard* to simulate, because evolving many forward rates jointly under one measure requires computing a genuinely complicated, correlation-driven drift at every time step of every path. This is the opposite trade-off from a Hull–White short-rate model, which is comparatively hard to calibrate cleanly to the smile but trivially easy to simulate (or even to grid/tree) because it has so few state variables. Which model a desk reaches for is, in large part, a bet on which of these two costs — messy calibration or expensive simulation — is more tolerable for the specific product being priced.

11.4 Where Part IV goes from here

Chapter 12 builds the change-of-numeraire machinery properly — the Girsanov-theorem toolkit needed to move a rate from its own measure to any other. Chapter 13 uses that toolkit to derive the BGM drift condition in full generality, for the spot, terminal, and any intermediate common measure. Chapters 14 and 15 then turn to the parameters themselves — what shape the instantaneous volatility and correlation functions should take, and how to extend the pure-lognormal postulate of this chapter to handle the smile and negative rates properly.

Chapter benchmark

Before moving on, you should be able to, without notes:

1. State BGM's central postulate: each forward rate is lognormal and driftless under its own forward measure.
2. Explain why this postulate makes BGM reproduce Black's caplet formula exactly, with no separate calibration needed for a single caplet.
3. Explain why evolving several forward rates jointly under one common measure forces every rate but one to acquire a nonzero drift, and why that drift cannot simply be assumed to be zero.